CS383 · SOFTWARE ENGINEERING

# Lecture 2 Introduction (II)

Midterm summary

**Muharib**

## What's inside


01 The four activities: Process activities

02 Software engineering and the Web

03 Software as a Service (SaaS)

04 SE Ethics: the eight principles

05 How does it appear in the midterm?

06 Common mistakes

07 Test yourself

## 01 The four activities: Process activities

Every software building process has four fundamental activities:

| # | Activity | What happens in it |
|---|---|---|
| 1 | **Software specification** | We define what software will be built and the constraints on its operation. |
| 2 | **Software development** | We design the software and implement it according to the specification. |
| 3 | **Software verification and validation** | We make sure the software is what the customer needs. |
| 4 | **Software evolution** | We modify the software so it keeps up with changes in the customer and the market. |

**Appeared in the midterm**

A question asked you to write the four activities in their logical order (2 marks).

## 02 Software engineering and the Web

In the last twenty years, systems have come to be built on the web and opened from the browser. So the software does not need to be installed on the device, and anyone in the

world can reach it right away.

### What did the developer gain?

```
Faster software delivery
```

```
Cheaper to change and upgrade
```

```
Reduce cost
```

### What changed in the way of developing?

- **Software reuse** became essential and important.
- Specifying all the requirements in advance is no longer practical, so we use **Incremental development**.
- The drawback: the user interface is limited by the capabilities of the browser.

## 03 Software as a Service (SaaS)

Instead of installing the whole program on the device, we take it as a service. The program runs on a cloud and we reach it through the web.

| Term | What it provides |
| --- | --- |
| **Platform** | Cloud components and a framework for building programs. It frees you from worrying about the operating system and storage. |
| **Infrastructure** | Scalable resources: servers, network, storage, provided through virtualization. |

| Benefits | Drawbacks |
| --- | --- |
| **Pay as you go**<br>You pay for what you use, monthly or yearly. | **Lack of control**<br>You have little control over it. |
| **Accessible anywhere**<br>You can reach it from any place. | **Vulnerable: security and trust**<br>It is exposed to security and trust problems. |
| **No need for expensive computers**<br>A cheap device is enough. | **Limited range of applications**<br>Only a few applications are available. |
| | **Internet dependent**<br>It does not work without the internet. |
| | **Performance issues**<br>It can be slow. |

**Appeared in the midterm**

An MCQ question about one of the drawbacks of SaaS. The correct answer was:

> Security/trust vulnerabilities and internet dependence

## 04 SE Ethics: the eight principles

The **IEEE/ACM Code of Ethics** has eight principles. In the exam you are given a description

and you write the name of the principle.

| Principle | Meaning | Words that point to it |
|---|---|---|
| **PUBLIC** | Act in a way that is consistent with the public interest. | responsibility for their own work |
| **CLIENT AND EMPLOYER** | Serve the interest of the client and the employer, without going against the public interest. | confidential information |
| **PRODUCT** | The product and its modifications meet the highest professional standards. | appropriate method, testing, integrity of data |
| **JUDGMENT** | Integrity and independence in professional judgment. | conflicts of interest |
| **MANAGEMENT** | Managers support the ethical management of development and maintenance. | informed of standards |
| **PROFESSION** | Raise the integrity and reputation of the profession. | reputation of the profession |
| **COLLEAGUES** | Be fair to your colleagues and support them. | fair hearing, colleague |
| **SELF** | Keep learning throughout your life and follow the ethics personally. | improve their knowledge, prejudices |

The lecture file I have is 8 pages and does not include the ethics slides, so the explanation here comes from the text of the Code itself. Check it against your instructor's slides.

## 05 How does it appear in the midterm?

The ethics question appeared in the second midterm for 6 marks. Here are its items and their answers:

| Description | Principle |
|---|---|
| Give a fair hearing to the opinions, concerns, or complaints of a colleague. | **COLLEAGUES** |
| Improve their knowledge of this Code, its interpretation, and its application to their work. | **SELF** |
| Ensure that software engineers are informed of standards before being held to them. | **MANAGEMENT** |
| Ensure an appropriate method is used for any project on which they work or propose to work. | **PRODUCT** |
| Disclose to all concerned parties those conflicts of interest that cannot reasonably be avoided or escaped. | **JUDGMENT** |
| Maintain the integrity of data, being sensitive to outdated or flawed occurrences. | **PRODUCT** |
| Accept full responsibility for their own work. | **PUBLIC** |
| Not give unfair treatment to anyone because of any irrelevant prejudices. | **SELF** |

**Watch out for these two**

The student in the graded paper got two items wrong: conflicts of interest is answered with **JUDGMENT**, and integrity of data is answered with **PRODUCT**.

Two midterms are a small sample. They show you the style of the questions, but they do not guarantee what the next midterm will look like.

## 06 Common mistakes

| Mistake | Correct |
| --- | --- |
| Putting conflicts of interest under Management | It is under **JUDGMENT**. |
| Putting integrity of data under Profession | It is under **PRODUCT**. |
| Saying SaaS gives you full control | The opposite, **Lack of control** is one of its drawbacks. |
| Reversing the order of the four activities | Specification, then Development, then V&V, then Evolution. |

## 07 Test yourself

1. List two benefits and two drawbacks of SaaS.
2. In which activity do we make sure the software is what the customer needs?
3. Which principle covers: **Keep private any confidential information**?
4. Which principle covers: **Ensure adequate testing, debugging, and review of software**?
5. Which principle covers: **Assist colleagues in professional development**?

## Answers

| # | Answer |
|---|---|
| 1 | Two benefits: Pay as you go, Accessible anywhere. Two drawbacks: Lack of control, Internet dependent. |
| 2 | **Software verification and validation**. |
| 3 | **CLIENT AND EMPLOYER**. |
| 4 | **PRODUCT**. |
| 5 | **COLLEAGUES**. |

The Test yourself questions were written by me. They are not from past midterms.